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Training youth for the labor market: Insights from an RCT in four African countries

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Training youth for the labor market: Insights from an RCT in four African countries

June 10, 2025

Abstract

We run an RCT in four African countries for the impact evaluation of a youth training program in a particularly vulnerable population. We complement self-reported measures on employment and job search with a real job opportunity, measuring participants' capabilities to seize opportunities in an incentivized setting. An analysis of applicants' CVs with AI techniques will shed light on the mechanisms behind improved labor market outcomes, by separately identifying skill acquisition from improved reporting. Our multi country study will allow us to understand contextual constraints that affect program success.

Keywords: training program, hard outcomes, youth unemployment, RCT

JEL Codes: I25, J24, O15

Study pre-registration: AsPredicted #202117

Proposed timeline: endline data to be collected from April to June 2025.

1 Introduction

Many governments and global NGOs implement youth training programs with the aim of facilitating the transition of young individuals into the labor market. This may be particularly important and useful for very vulnerable populations with a lack of skills and little social support (Carranza and McKenzie, 2024). Yet, the evidence on the effectiveness of such programs is mixed, but it suggests that its effects depend heavily on the specific implementation characteristics, program features and other contextual effects (Honorati and McArdle, 2013).

Youth training programs are nowadays usually designed in such a way that they can have a positive influence on employment and earnings by combining program characteristics that have proven beneficial (Zimmermann et al., 2013). The YouthCan! program, implemented by the NGO *SOS Children's Villages* (SOS CV) in over 50 countries, is a good example since it incorporates training on basic professional skills (like CV-writing, where to find jobs or how to prepare for an interview) and on-the-job training for practical skills. If possible, it complements these trainings with one-to-one mentoring. Internal evaluations find a positive effect of the program, with 27% of program participants finding a job after the training and up to 38% for those aged 22 or older (SOS Children's Villages International, 2024). Yet, these effects may be biased due to the self-selection of participants into the program and into the evaluation.

We are currently running a Randomized Control Trial (RCT) to evaluate this program in four African countries: Benin, Cape Verde, Rwanda, and South Africa, all with a large percentage of their population living in financial deprivation. The first contribution of our multi-country approach is therefore to investigate the effectiveness of youth training programs for very vulnerable populations by considering different contexts. Our sample is particularly vulnerable, with 67% of youth receiving direct support from SOS CV. These supported youth have smaller professional networks, lower self-esteem and lower confidence in their professional skills (at baseline) compared to the already vulnerable youth living in the communities where SOS CV are located (the remaining 33% of our sample). Moreover, to the best of our knowledge, our evaluation will be the first to provide evidence from a multi-country study testing a highly standardized intervention, which can help us understand the role of local labor market conditions and economic growth on the success of these programs. This way, we will complement the

evidence about the effects of macroeconomic contexts, which, so far, stems exclusively from developed countries with their very different labor markets[1].

Our second contribution to the literature is to examine in more detail job seekers' job search strategies by analyzing the application behavior and material of the youth in response to a real job offer. We will offer the youth in our study a short and very well-paid job taking place at their local SOS CV office. Participants will first need to express an interest in such a job and, after receiving further details, can submit a CV and motivation letter that we will analyze with Large Language Models (LLMs). This analysis will allow us to distinguish skill acquisition (since we know which skills were acquired during training) from skill reporting, two channels which potentially mediate treatment effects. We expect better quality CVs of youth in the treatment group, both by listing more skills and by providing a more complete report of central elements of any CV (like contact information, education, language skills, or work experience). These hard outcomes will also allow us to measure effects of the program while attenuating problems of reporting bias.

The third contribution to the literature comes from analyzing the effects of the program on the size of professional networks. This aspect is related to our focus on a particularly vulnerable population, but adds a further contribution by looking at how our intervention affects the networks of very disadvantaged individuals. So far, only a few studies have investigated the importance of training programs that provide connections to the private sector, again stemming exclusively from developed countries (Kashefpakdel and Percy, 2017; Mann et al., 2018; Musset and Kurekova, 2018). Networks are essential in job search, especially in developing countries (Wahba and Zenou, 2005; Berardi, 2013; Wheeler et al., 2022). Given the characteristics of the youth in our sample, most of whom do not live with their families or are at risk of losing parental care, we expect an increase in their professional networks, which can be one of the mechanisms behind the treatment effects on employment.

The main outcome of the evaluation, given the aim of the program, will be whether the youth are employed or not a couple of months after the training. While the latter will rely on self-reports, we will also take as a main outcome the application rates for the

[1]This evidence comes from longitudinal analysis of a single country (Lechner and Wunsch, 2009; Forslund et al., 2011), from pseudo-panels (Kluve, 2010) or from a meta analysis (Card et al., 2018).

temporary job that we will offer and, conditional on applying, the quality of their CVs. As secondary outcomes, we will measure the submission of a motivation letter, changes in the size of their professional networks, and more standard labor market variables such as working hours, work contracts, and labor income from their self-reported jobs. Heterogeneity analysis on the level of vulnerability (supported or not by SOS CV) and on labor market conditions made possible by our multi-country approach and by considering different locations within our four countries, will provide valuable insights on the relevance of contextual effects for the effectiveness of these programs.

2 Research Context

We evaluate the labor market training program *YouthCan!* already implemented by the NGO *SOS Children's Villages* in more than 50 countries. Our study takes place in four African countries: Benin, Cape Verde, Rwanda, and South Africa, all of which have more than 35% of their population under the (respective) national poverty line (World Bank, 2024). Other than that, these countries speak different languages and are far apart geographically. They are also quite different regarding GDP, GDP growth, employment, and informality in the labor market (see Table 8 in Appendix C), allowing us to evaluate the training program under very different conditions.

2.1 SOS Children's Villages

SOS Children's Villages (SOS CV) has been founded as a non-governmental organization in 1949 to help children who lost their parents during World War II. Today, SOS CV works in more than 130 countries, supporting children without parental care and families at risk. It has two basic programs for child care: "Family Strengthening" (FS) and "Alternative Care" (AC). The FS program helps families at risk to prevent children from losing parental care, providing close supervision to make sure the children are well protected. The AC program is for children for whom staying in the family is not possible. The children live in an alternative family with other children and a caregiver who provides professional care and education.

SOS CV aims at helping children and youth leaving AC or FS programs to become self-reliant and financially independent, offering employability and entrepreneurship

programs. These programs are also provided to vulnerable youth in the communities where SOS CV are located.

2.2 YouthCan!

YouthCan! is a job training program by *SOS Children's Villages* for young people about to enter the labor market. It was first implemented in 2017. In 2023, 17,800 youth participated worldwide (SOS Children's Villages International, 2024). Most participants are between 15 and 25 years old and are part of the FS or AC programs. A smaller part are interested youth from the communities where SOS CV centers are located. The training is offered by volunteers from partner organizations such as DHL, Allianz, and Johnson & Johnson. The program offers life and technical skills training, practical work experiences, and mentoring. Youth learn skills such as CV-writing, how to apply for jobs and prepare for interviews, and participate in specific skills workshops like computer programming. A detailed description of the activities included in the training of our evaluation is presented in section 3.2.

3 Research Design

We evaluate this training program in four African countries through a randomized control trial. The eligible youth for the study are between 16 and 25 years of age, have not participated in *YouthCan!* training before but are interested in participating, and are selected by SOS CV (see section 4.1).

3.1 Hypotheses, Research Question and Outcomes

The main research question is if and to what degree do youth training programs increase employability of vulnerable populations.

We hypothesize that youth in the training group are more likely to get a job, more likely to apply for jobs and to write higher quality CVs. We also expect differences by contextual effects, with larger treatment effects for youth supported by SOS (more vulnerable) and larger effects in countries with lower educational levels and favourable macroeconomic conditions (see section 5.5). Finally, we hypothesize that youth in the

treatment group are more likely to have higher income and have larger professional networks.

An important contribution of our study is to understand the effects of this type of training on particularly vulnerable populations. We do so in our heterogeneity analysis (see section 5.5) by comparing youth supported by SOS CV to those who live in communities where SOS CV are located. Section 4.4 shows that youth supported by SOS have smaller professional networks and perceive themselves as having fewer professional skills. The training specifically addresses these challenges with the professional skills training and by connecting the youth with the volunteers who provide the training. Additionally, SOS-supported youth have lower self-esteem and lower patience, both of which are predictors of job search and job performance (Drago, 2011; Magnusson and Nermo, 2018; DellaVigna and Paserman, 2005; Burks et al., 2009). Importantly, lower self-esteem is not unique to our sample but it is common in children from unstable families, such as orphans (comparable to our AC sample, e.g., De Weerdt et al., 2017) and families with conflict or single-parent families (comparable to our FS sample, e.g., Musick and Meier, 2010; Lahiri and Verma, 2023). The training can address these challenges indirectly and we will measure changes in these dimensions as potential mechanisms for the effects.

Given the setup of our study we can also contribute to the understanding of the role of labor market conditions on the success of these programs. Tables 8 and 9 in the appendix show country- and location-level indicators for the ease with which youth can find a job. We hypothesize larger treatment effects in locations with a potentially larger labor market and with more favourable macroeconomic conditions. We also hypothesize that macroeconomic and institutional indicators are more relevant than location indicators. A discussion is presented in section 5.5.

Our main outcome variable, employment, as well as secondary outcomes on network size and employment conditions will be self-reported at the endline survey. Given the evident objective of the training program, one might be concerned of experimenter demand effects, social desirability bias, or even gratitude towards the program and SOS CV to affect the responses to questions on job search and effort (Dupas and Miguel, 2017; De Quidt et al., 2018; Dhar et al., 2022). To address this concern, we first include questions of the social desirability scale (Crowne and Marlowe, 1960), and

we complement our research design by measuring hard outcomes.

To do so, we will offer a real (temporary) job (in a similar way to Flory et al., 2015; Abel et al., 2019, etc.) with a very attractive wage at the end of the endline survey. To maximize the number of applicants, the tasks listed in the job ad will vary from very simple ones–such as ordering paperwork– to more complex ones–involving computer skills. This will incentivize the youth to list all of their skills on their CVs. We will offer an hourly wage following European standards, which will be financially very attractive compared to the local labor market. The time effort will be low, and the candidate will be able to select the day and time to do the task(s). In this way, we do not harm the youth by preventing them from getting (or continuing in) jobs in their local labor market. We will first collect their expression of interest (separately from the actual application and unaffected by treatment) in order to differentiate between general interest in jobs and the willingness to invest time and effort into applying. The exact procedure is presented in section 4.2, while the job ad and the text for the expression of interest are presented in Appendix B.

As hard outcomes, we will measure application rates and the quality of their CVs (going beyond Abel et al., 2019, who only look at the former). There should be no difference in just *expressing interest* in the job between treatment and control group, since this job opportunity will be unforseeable for participants (implying that those in the treatment group will not be able to prepare for this in advance) and will be non-specific[2] (implying that different reservation wages should not have an impact, in particular because our job will pay very well).

Differences in *application rates* could already be attributed to the training. CV writing is a skill that is specifically targeted by the training program, apart from being a critical skill in any labor market search. We hypothesize that individuals in the treatment group will apply at higher rates since the cost of preparing a CV should be lower, their motivation could be higher, and their confidence in applying successfully might be stronger. We also expect better quality CVs controlling for employee quality (i.e., skills acquired during their training), meaning that we will be able to distinguish better CVs due to skill acquisition vs. improved reporting. Details on the use of Artificial Intelligence for this purpose are presented in section 5.2.

[2]The exact wording can be found in Appendix B.

The submission of a motivation letter for our temporary job will be voluntary, and we will take this as a secondary outcome since it can be interpreted as higher levels of motivation and as an increased capability of seizing opportunities.

The additional secondary outcomes will be collected at the endline survey. We will consider working hours, work contracts, labor income (from their self-reported job), and changes in the size of their professional network. Network effects have been hypothesized as a relevant mechanism in these types of training programs (Alzúa et al., 2016; Maitra and Mani, 2017), since they are very relevant during job search, especially in developing countries (Wahba and Zenou, 2005; Berardi, 2013; Wheeler et al., 2022). Additionally, building professional networks is a specific goal of the *YouthCan!* program, and they can be particularly important in our context, considering the social fragility of our participants.

Mentoring could be very relevant given the vulnerability of our population, not only through providing emotional support and functioning as role models, but also by compensating for small professional networks. We do not put too much emphasis on this mechanism given the small sample size of mentored participants. Nevertheless, to gain qualitative insights, we will include in the endline survey several questions to explore these mechanisms.

The methods used for this analysis and the specific survey questions are presented in sections 5.1 and Appendix A, respectively.

3.2 Intervention

The intervention is designed to evaluate the *YouthCan!* training program. For this, we randomly distributed interested youth satisfying the participation requirements into control, treatment, and intensive treatment groups. The control group does not participate in any training of *YouthCan!* during 2024, but they were promised to receive the training during 2025 (after our endline survey). The treatment group takes part in three activities:

- Career guidance and orientation on job profile (8 hours): youth receive information about different jobs and their requirements. Participants should define which jobs fit well with their interests and skills.

- Basic skills for employment (8 hours): youth learn how to write an application, a CV, and a motivation letter and how to prepare for a job interview.
- Professional skills training (40 hours): youth receive specific practical training such as painter training, typing courses, computer programming, or English language course. Participants can choose which specific skill they want to learn.[3]

All these activities take place in groups. Participants assigned to the intensive treatment group take part in these activities and additionally receive one-to-one mentoring sessions (8 hours) with volunteers from SOS CV's partner organizations (for example DHL). This training is adjusted to the needs of the youth and guides them through their job search. In total, the treatment group receives 56 hours of training, and the intensive treatment group receives 64 hours.

3.3 Sample size and statistical power

We aimed at a sample size of 600 youth, considering which countries–already implementing the program–would be willing and able to collaborate with us in the impact evaluation. With this number in mind, we agreed that four countries would each provide training to approximately 75 young people. The four participating countries are illustrative of Sub-Saharan Africa.

In total, 300 youth were assigned to the control group, 220 to the treatment group, and 78 to the intensive treatment group. We will pool the two treatment groups and only consider the intensive treatment group for heterogeneity analysis.[4] Thus, our power calculations assume one control group with 300 observations and one treatment group with 298 observations.

We calculated the power of our experiment to detect the effect of whether the proportion of people having a job is larger in the treatment than in the control group. When performing a one-sided t-test with our baseline sample size, we are able to detect an increase of 0.203 standard deviations (with a significance level of 5%) with a probability of 80%.

[3]The offer of the skills taught depends mainly on the availability of volunteers. Youth choose which skills they want to learn among the possibilities offered.

[4]The distribution of youth in control, treatment, and intensive treatment groups per country can be found in Table 6 in Appendix C.

Table 1: Number of Observations by Country

Country	Control	Treatment	N
Benin	71	69	140
Cape Verde	74	72	146
Rwanda	73	75	148
South Africa	82	82	164
Total	**300**	**298**	**598**

Notes: Number of observations per country in control and treatment group at baseline.

From our baseline data, we observe that, on average, 15.7% of the youth have a job at baseline. The standard deviation is 0.364. When we account for the stratification we used during our randomization process, the unexplained variance of the baseline outcome variable decreases, and the unexplained standard deviation of the outcome is 0.343.[5]

This corresponds to a minimum detectable effect of 6.97 percentage points in the treatment group across all countries. In our setting, this would mean that around 22.7% of youth in the treatment group have a job at the endline survey, while the proportion in the control group stays at around 15.7%. This 44% increase in employment is high but attainable, as shown by other interventions similar to ours (Das, 2021). Additionally, we can expect larger effects since our baseline levels of employment are low, our sample is more vulnerable (as seen in Table 5), our training incorporates job application skills (proven successful in Abebe et al., 2020) and the training also provides youth with points of contact in the private sector (proven successful in developed countries, in e.g. Mann et al., 2018).

We expect the take-up to be high: youth declared their interest in the training program before being invited to the baseline survey, and we excluded youth from our study that indicated not to be interested in any activity of *YouthCan!*

We expect a low attrition rate since youth in our study were already in contact with SOS CV and should be easy to reach for the follow-up survey and willing to participate again.

[5]We regress the baseline outcome variable on strata dummies. The remaining variance of the regression is the variance in the outcome variable that is not explained by the variation between strata.

In the meta-analysis of labor market training by McKenzie (2017), attrition in the studied papers varies between 17% and 46%. We calculate the power of our experiment for different attrition levels and a minimum detectable effect of 6.97 percentage points in table 2.

Table 2: Power for different attrition rates

	Attrition			
	0%	10%	20%	30%
Power	0.8	0.762	0.719	0.668

Notes: N_{total}=598, power is shown for a minimum detectable effect of 0.203 standard deviations using a one-sided t-test for a positive treatment effect. The sd for baseline outcome variable "currently having a job" is 0.364. When accounting for stratification the sd for the unexplained variation is 0.343. The minimum detectable effect is 6.97 percentage points.

4 Data

The main data for our analysis is collected by us through online surveys[6] and an online job offer. The data is obtained from a selection of youth who satisfy the participation criteria and were assigned to treatment or control groups after the baseline data collection.

4.1 Recruitment

Participants for the study were recruited through a combination of targeted outreach and community-based approaches. Local coordinators reached out to youth that meet our study criteria (between 16 and 25 years old, interested in participating in *YouthCan!* and who did not participate before) via different channels:

For youth directly affiliated with SOS CV, local coordinators selected individuals who met the study criteria and reached out to them through direct communication. This included in-person discussions, informational text messages via email or phone, and promoting the opportunity of participation in *YouthCan!* during information sessions.

For youth not supported by SOS CV, recruitment was conducted in collaboration with local networks, such as community-based organizations, which helped identify

[6]We use the oTree platform (Chen et al., 2016) for programming and conducting the online surveys.

suitable candidates. Digital communication channels, including phone and email, were used to reach potential participants. Additionally, in-person information sessions were organized to introduce the program.

4.2 Data collection and processing

The baseline survey was completed in computer labs in the training centers of SOS CV, at the respective program locations. The local team in each of the four countries invited eligible youth to the center's computer lab on a specific date, where they conducted survey sessions with about 15 youth per session. The survey was available in English, French, Portuguese, Zulu, and Sepedi to maximize its comprehension among the youth in each of the participating countries. Additionally, local supervisors were present in the computer lab to help in case of interpretation/comprehension issues and to ensure that youth answered the survey independently. Note that supervisors were SOS CV employees. Given that they were supervising both the control and the treatment group and given that they could not observe the survey questions or the answers, we are confident that this did not introduce any bias.

Each survey participant received a lump-sum payment covering their travel costs, a lump-sum payment for their participation, and an individual payment based on their decisions in four tasks regarding time preferences, risk preferences, and altruism. The magnitude of these payments was based on national minimum or average wages. On average, survey participants received 17$ PPP[7] to answer the baseline survey, including individual payment and travel costs. The median participant took 47 minutes to complete the survey.

The same approach will be followed for the endline data collection.

For the measurement of hard outcomes, we will offer a temporary job in each location as follows:

- At the end of the endline survey, we inform participants about the available job and a basic description of its characteristics.
- Participants can express their interest.
- Interested participants receive more details on the job.

[7] Purchasing-Power-Parity index for 2023.

- Participants apply via email.
- We select up to three participants per country who get the job.

We encourage youth to take a photo of the online job ad to ensure that youth who do not have an email account have time to create one or send us their application via another account. The job advertisement can be found in Appendix B.

4.3 Randomization and balancedness

After baseline data collection, we randomly allocated youth into three groups: control, treatment, and intensive treatment. In each country, about 75 participants are part of the control group, 55 of the treatment group, and 20 of the intensive treatment group. To take into account geographical differences in the labor market, we stratified the randomization by program location. The strata are then 3 program locations in Rwanda, 3 in Benin, 2 in Cape Verde and 7 in South Africa, for a total of 15 strata. We combined two geographically close locations in South Africa in one strata to ensure a reasonable sample size.[8]

We applied the minMSE-algorithm (developed by Schneider and Schlather, 2017) for each stratum. This algorithm minimizes the differences between the three groups on given covariates. Using this algorithm, we produced 100 treatment allocations and randomly picked one allocation for randomization, leaving the remaining 99 allocation vectors for randomization inference (see section 5.1). The advantage of this algorithm is that it produces balanced allocations even for smaller samples and many covariates. Table 3 presents important covariates and statistics about the differences between treatment and control groups. See Table 7 for a complete list of the variables considered for the randomization procedure.

4.4 Descriptive statistics

In Table 4, we show the prevalence of certain characteristics in our sample. 50% of our participants are very motivated to find a job, but only 28% feel well prepared to apply for a job and only 23% are convinced that they have the necessary professional skills. Correspondingly, 50% of our participants are interested in joining the professional skills

[8] We have a minimum of 14 observations per strata.

Table 3: Balancedness Table

Variable	Levels	Control	Treatment	Difference	p-value
Age	16 - 25	20.233	20.332	0.099	0.627
Male	yes/no	0.457	0.490	0.033	0.416
Married	yes/no	0.017	0.040	0.024	0.083
Has kids	yes/no	0.167	0.188	0.021	0.497
Has high school edu.[a]	yes/no	0.710	0.688	-0.022	0.557
Has a job	yes/no	0.163	0.151	-0.012	0.679
Looking for a job	yes/no	0.757	0.742	-0.015	0.672
Monthly income[b]	in $ PPP	21.664	20.675	-0.988	0.842
Has a smartphone	yes/no	0.590	0.624	0.034	0.393
Network index	1 (low) - 5 (high)	3.276	3.243	-0.033	0.664
Supported by SOS CV	yes/no	0.667	0.678	0.011	0.771

Notes: Means and tests for differences between treatment and control group. 300 people are in the control and 298 in the treatment group. a) 25 youth (4% of the whole sample) have no high school degree, are below 18 and are still in school. b) Most participants (69%) have no income.

training.

Table 4 shows that 41% of the youth in our sample had previously been employed in any capacity. Among them, almost half had only unpaid internships, while only 22.7% of the whole sample had a paid job, which also includes paid internships.

At baseline, an average of 15.7% of the youth in our sample had a job. Of these, one-third were doing unpaid internships, while one-fifth of them were in paid internships. Only 7% of the total sample had a paid job that is not an internship. This low proportion of youth with paid jobs, combined with the high rate of job-seeking (see table 3), highlights the need for targeted labor market training for this group.

Most of our sample (58%) comes from families that are part of the Family Strengthening program. Additionally, 9% live in Alternative Care or shared homes supported by SOS CV. The vulnerability in their family backgrounds makes them a key target group for support initiatives to improve their employment opportunities.

Table 5 shows different vulnerability criteria of the youth supported by SOS CV (roughly two thirds of our sample) compared to youth participating in our evaluation but who reside in the communities where SOS CV centers are located, and who

Table 4: Labor market experience and SOS program

I am motivated to find a job. (Strongly) agree	49.8%
I am prepared to look and apply for a job. (Strongly) agree	28.1%
I have the necessary professional skills that can help me find a job. (Strongly) agree	23.1%
Interested in professional skills training.	48.5%
Ever had a job	41.3%
- At least one paid job	22.7%
Has a job now	15.7%
- Unpaid internship	5.7%
- Paid internship	3%
- Own business	2.3%
Receives SOS support	67.2%
- Part of FS program	58%
- AC or supported	9.2%

Notes: Descriptive statistics of our study sample, $N = 598$. The first three statements were answered on a 5-point Likert scale from "Strongly disagree" to "Strongly agree".

are not supported by SOS CV. Compared to community youth, supported youth have smaller professional networks, tend to be less patient and report lower motivation and self-esteem. Importantly, they perceive themselves as having fewer professional skills, making them a particularly relevant group for labor market interventions that not only enhance their skills but also boost their confidence in their own abilities. In this context, direct contact with volunteers from firms could play a crucial role in bridging their network gap, providing connections that facilitate labor market entry.

The differences presented in Table 5 are striking, especially considering that community youth are also disadvantaged. SOS CV centers are usually located in disadvantaged suburban areas where there is a high prevalence of social and economic problems which require their intervention[9].

[9]For example, SOS CV Rwanda explains their choice of location in Kayonza as follows "It started in

Table 5: Differences between youth supported and not supported by SOS

Variable	**Levels**	**Supported - Not-supported**	**p-value**
Male	yes/no	0.071	0.173
Single caregiver	yes/no	-0.015	0.753
Has high school edu.	yes/no	-0.056	0.180
First caregiver has hs edu.	yes/no	-0.125	0.011
Ever had a job	yes/no	0.048	0.238
Looking for a job	yes/no	0.034	0.429
Monthly income	in $ PPP	-4.656	0.459
Has a smartphone	yes/no	-0.062	0.189
Network index	1 (low) - 5 (high)	-0.190	0.045
Motivation index	1 (low) - 5 (high)	-0.189	0.045
Prof.-skills index	1 (low) - 5 (high)	-0.222	0.021
Self-esteem index	1 (low) - 5 (high)	-0.113	0.063
Feeling depressed (Index)	0 (never) - 4 (always)	0.172	0.115
Aggr. time pref. index[a]	0 (impatient) - 20 (patient)	-1.362	0.023

Notes: Difference between youth supported by SOS and youth not supported by SOS. The difference is adjusted by country-level differences and age. The underlying regression is $outcome_i = \beta supported_i + \delta age_i + \gamma country_c$. β is the coefficient for the difference displayed in the table. N=196 for community youth and N=402 for SOS supported youth. a) Combines answers of trade-off between "now" and "in one week" and between "in one week" and "in two weeks".

The characteristics of our sample will allow us to understand the potentially different effects of youth training programs depending on different levels and types of vulnerabilities.

September 2010 to respond to the alarming problems of children such as orphanage, abandonment, and separation of children from their families. The later effects resulted from polygamy, high prevalence of HIV/AIDS, family conflicts, a big number of children per household, poverty, and unemployment, migration to other provinces and neighboring countries in the region, etc." (https://www.soscv-rwanda.org.rw/-our-villages-.html)

4.5 Timeline

The baseline survey was conducted between May and July 2024 in South Africa, Benin, and Rwanda and between September and November 2024 in Cape Verde. Activities started around two months after the baseline survey was completed. The order of activities is determined by the availability of the local teams and volunteers, making sure that they fulfill the agreed number of training hours. Benin and South Africa finished the activities at the end of December and Rwanda and Cape Verde will finish them in April 2025.

We plan to do the endline survey in mid April 2025 in South Africa, mid May 2025 in Benin and end of June 2025 in Rwanda and Cape Verde. This means that the endline survey is run in all countries between 10 and 14 weeks after the end of the activities. Youth will have a two-week time limit after the endline survey to apply for the temporary job.

5 Analysis

5.1 Statistical methods

We will estimate the following regression to measure the average treatment effects of the program:

$$y_i = \beta T_i + \delta y_{i0} + \gamma_s + \epsilon_i$$

Where y_i is the outcome of respondent i, T_i equals 1 if respondent i is in the treatment group and 0 otherwise, and y_{i0} is the outcome at baseline (if available). γ_s are strata fixed effects. We use standard OLS regressions with heteroscedasticity-robust standard errors.

When the outcome variable is an index (such as the measurement of professional networks) we combine the answers of several survey questions into this index by using weights from a principal-component analysis.

To back up our results, we will use randomization inference. During the randomization procedure using the minMSE algorithm, 100 balanced possible treatment assignments

were created. We picked one of them for our actual treatment group allocation. The other 99 treatment vectors will be used for randomization inference.

For the heterogeneity analysis proposed in section 5.5, we will use OLS regressions with interaction terms for the selected variables. Additionally, we will use Causal Forests, developed by Wager and Athey (2018), to back up our results and to analyze any heterogeneous treatment effects beyond the pre-specified ones. To look at heterogeneity by market size, we will remove γ_s and include country fixed effects instead.

We will test for differential attrition by regressing a dummy for follow-up participation on treatment status. If we detect non-random attrition, we will report upper and lower bounds of the treatment effects following the approach proposed by Lee (2009).

5.2 Evaluation of CVs

We use large language models (LLMs) to systematically assess CV characteristics. To ensure accuracy, we validate the procedure through human review and refinement. LLMs extract key characteristics and compile them into a structured summary of all CVs.

Our approach distinguishes between two dimensions of CV quality: reporting quality and skill acquisition.

The *reporting quality* category includes the following elements:

- Inclusion of contact information (email address or phone number),
- Reporting educational background (institution or degree),
- Reporting of language skills,
- Reporting of internships or work experience prior to the training.

Possessing these characteristics is independent of the treatment itself, as all participants have at least some relevant background (e.g., schooling, native language, prior work experience) and these are balanced across treatments. However, we hypothesize that participants in the treatment group, who received CV-writing training, are more likely to *report* these elements, reflecting improved reporting skills.

The *skills acquisition* category includes:

- Reporting of computer or digital skills,
- Reporting of organizational skills,
- Language proficiency levels,
- Reporting of internships or work experience acquired after the training.

We expect these skills to appear more frequently in the CVs of the treatment group, reflecting the content and focus of the training they received.

To synthesize the information and reduce concerns around multiple hypothesis testing, we construct two indices: a *reporting index* and a *skills index*, aggregating the respective binary indicators in each category. We will then regress these indices on treatment status to estimate the effect of the intervention.

The observed CV quality is conditional on application itself. To assess whether differences in CV quality indeed stem from the treatment rather than from differences in the composition of applicants, we test for any imbalances using baseline information between treatment and control applicants and include unbalanced variables as controls in our analysis of CV quality. This approach ensures that our estimated treatment effects on CV quality are not confounded by selection into application.

We hypothesize that treatment participants will have significantly higher values on both indices.

5.3 Data cleaning and sample restrictions

The baseline data collection continued until we had around 150 participants per country who satisfied our participation criteria. This means that interested subjects outside of the age range 16-25, those who had already participated in *YouthCan!* activities and those not showing interest in participation were not admitted to the program.

We also excluded outliers in income distribution at baseline. By excluding the top 3% of the income distribution at baseline we could reduce the standard deviation of monthly income by around 50% (most of the youth from our study do not have any income). This exclusion increases the power of our study regarding monthly income. Additionally, it seems likely that the income questions were not answered correctly

by these outliers and could not be trusted, which would complicate a proper impact evaluation. Treatment assignment was performed after these exclusions.

If there are missing values, we will follow the standard approach by imputing their median value at the country level and include an indicator variable for these imputations (McKenzie, 2022).

5.4 Multiple outcome and multiple hypothesis testing

In our main analysis, we study whether the treatment group is more likely to have a job. We also examine whether people from the treatment group apply to our attractive (and temporary) job offer at higher rates. Additionally, we study the qualitative dimension of these two outcomes: 1) we look for treatment effects on working hours, labor income and formal employment (i.e. whether there is a work contract). 2) We analyze the quality of the CVs and whether they hand in a motivation letter when applying for our temporary job. Within these two categories, we will adjust the p-values for multiple hypotheses testing using the method by Romano and Wolf (2005).

We will also test whether the treatment group built a larger professional network. We ask three network-related questions that can be found in Appendix A. We aggregate these questions into a network index, calculated as a weighted average. The weights are derived using the first principal component obtained from a principal component analysis (PCA). The PCA is done separately for each country to adjust for contextual differences [10].

Other variables of interest, mainly to understand the mechanisms behind the effects, are optimism, motivation, and self-confidence. We will combine them in a single index using different survey questions (presented in Appendix A).

5.5 Heterogeneous Effects

We will analyze heterogeneous treatment effects by an objective measure of extreme vulnerability (receiving support from SOS CV), and by macroeconomic/labor market conditions (which should account for important differences between countries and loca-

[10] The weights are similar across countries, giving around 20% to the first question and 40% to each of the last two questions.

tions). Additionally, we analyze heterogeneous effects by treatment intensity (whether the treatment includes mentoring), gender and baseline levels of education.

An important variable by which we expect heterogeneous treatment effects is our measure of vulnerability. Here we differentiate between participants who are officially supported by SOS CV (through family strengthening or alternative care) and those who are not. We hypothesize stronger effects for participants who receive support from SOS CV (in contrast with youth from the community, as explained in section 2) since they have several disadvantages, such as smaller professional networks, lower self-esteem and patience and lower professional skills at baseline (Table 5). Differences in the treatment effect would highlight the importance of different degrees of vulnerability of the beneficiaries for training programs.

Labor market conditions will determine the ease with which participants can find a job after participating in the program, with vibrant economies offering more employment opportunities to the youth. The question is whether youth who participated in our training can profit from good macroeconomic conditions to a higher degree than those in the control group. In their review of active labor market policies, Card et al. (2018) find evidence suggesting that these trainings work better in recessionary markets. This evidence, however, comes mainly from developed countries[11], which are not comparable to the labor markets of the countries we analyze.

Table 8 in Appendix C shows economic indicators per country. We expect higher employment levels at endline in Benin and Rwanda, which have low youth unemployment, high informality (which simplifies getting a job or starting a business) and high projected GDP growth. We expect higher treatment effects in these same countries since they also have lower levels of education, and our training program could be addressing the lack of skills in labor supply. If our hypothesis doesn't prove right, we could conclude that the same mechanisms driving the results of Card et al. (2018) are in place in developing settings.

We can, however, take a closer look at the heterogeneous effects of the treatment by looking at characteristics of each location in which the program is implemented. Table 9 shows indicators on the potential size of the labor market (proxied by the population size of the town or city where the SOS CV center is located, the distance to the closest

[11] They include program results from Denmark, France, Germany and the United States

large city and the number of restaurants), with larger labor markets offering more opportunities to the youth. For example, we expect larger effects in Kigali, in Rwanda, than in Dassa-Zoumé in Benin, even if the macroeconomic conditions in both countries are good. We also expect the macroeconomic and institutional conditions to be more relevant for the success of the program: we expect lower effects in Rustenburg, in South Africa, even if it has a very large potential labor market. These hypotheses are consistent with employment data from our sample at baseline, where the employment rate in Kigali, Dassa-Zoumé and Rustenburg was of 33%, 7%, and 0% respectively.

An important objective of this impact evaluation is to understand if mentoring, a resource-intensive component of the *YouthCan!* program, impacts employability. Indeed, mentoring could have a similar impact as job search assistance, which has proven more successful for disadvantaged participants (Card et al., 2018). Unfortunately, given the complexity of finding one-to-one mentors for participants, this additional component of the treatment could be assigned to 20 participants per country only. We, therefore, analyze differences stemming from mentoring only as part of the heterogeneity analysis and not as a main outcome. We hypothesize stronger positive effects for youth who were assigned a mentor.

Previous literature has identified differences by gender and baseline levels of education in the effectiveness of employability programs. Monk et al. (2008) find that the effectiveness of training depends heavily on whether participants have formal education or not. Regarding gender, Attanasio et al. (2011), Blattman and Ralston (2015), Chakravarty et al. (2019) and Acevedo et al. (2020), for example, found stronger effects for women, while Cho et al. (2013) and Alzúa et al. (2016) found it for men. We abstain from hypothesizing the direction of this effect and any difference found will only be reported as additional evidence.

Finally, we will include questions from the Marlowe-Crowne social desirability scale (Crowne and Marlowe, 1960), which was shown useful in addressing experimenter demand effects (e.g. Dhar et al., 2022; Bahety et al., 2021). We will examine heterogeneity depending on whether respondents score high or low on this social desirability scale. This should also address any possible bias stemming from the presence of SOS CV employees as supervisors during the endline survey.

6 Administrative information

6.1 Funding

The activities that are part of the *YouthCan!* training program are funded by *SOS Children's Villages*. The impact evaluation is funded by the Max Planck Institute for Research on Collective Goods, Bonn. This includes payments and travel costs for participants in the surveys, as well as the wage for the temporary job we offer at endline.

6.2 Ethics Approval

Ethics approval for this study was obtained by the *Ethics Committee of the Faculty of Management, Economics and Social Sciences (ERC-FMES)*, University of Cologne (Reference: 240004JT).

6.3 Conflicts of interest

We have no conflicts of interest to declare.

Appendices

Appendix A Survey questions

A.1 Supported by SOS CV

What type of SOS programme are you part of?

- I live in an SOS family (in an SOS village or foster home)
- I live with my family and we are supported by SOS Children's Villages;
- I live alone or in a home with other young people, supported by SOS
- I live in a community where SOS Children's Villages is present
- None of the above options

A.2 Employment

Have you done any work or internships?

- Yes
- No

Are you still working in this job?

- Yes
- No

Please indicate the type of your last/ current job.

- Unpaid internship/apprenticeship
- Paid internship/apprenticeship
- Paid work with limited contract of less than 1 year
- Paid work with limited contract longer than 1 year
- Paid work with unlimited contract
- Paid work without a contract
- I started my own business
- Other (please specify): ...

A.3 Network Index

How many persons who hold steady jobs do you typically interact with?

- 0
- 1-2

- 3-5
- 6-9
- 10 or more

Please indicate how strongly you agree or disagree with the following statements:
(Scale from 1 - Strongly Disagree to 5 - Strongly Agree)

- I have a person (an adult, friend, or mentor) or group of people I regularly speak to for ideas, guidance, or help regarding education or work.
- I am connected with external professionals that can support me.

A.4 Motivation Index

Please indicate how strongly you agree or disagree with the following statements:
(Scale from 1 - Strongly Disagree to 5 - Strongly Agree)

- I am happy with my job and the activities I perform.
- I am confident about my ability to find a (new) job.
- I am motivated to find a (new) job.
- I am positive about my future.
- I have a plan for the next steps in my life.
- I am confident about my ability to start my own business.
- I am motivated to start my own business.

A.5 Professional Skills Index

Please indicate how strongly you agree or disagree with the following statements:
(Scale from 1 - Strongly Disagree to 5 - Strongly Agree)

- I know how to write a CV.
- I know how to prepare for a job interview.
- I am prepared to look and apply for jobs.
- I am prepared to start my own business.
- I have the necessary social skills to succeed in life (e.g., how to work in a team, how to communicate with others, etc).
- I have the necessary professional skills (like painting skills, computer/digital skills, project management, personal finance, agriculture training) that can help me find a job.
- I have the necessary professional skills (like painting skills, computer/digital skills, project management, personal finance, agriculture training) that can help me start my

own business.

- I can interact professionally in a work environment.

A.6 Depression Index

Please indicate how often you have been bothered by any of the following feelings over the last two weeks: *(Scale from 1 - Never, 2 - 1-3 days, 3 - 4-6 days, 4 - 7-10 days, 5 - 11-14 days)*

- Little interest or pleasure in doing things.
- Feeling down, depressed, or hopeless.

A.7 Self-esteem Index

Please indicate how strongly you agree or disagree with the following statements: *(Scale from 1 - Strongly Disagree to 5 - Strongly Agree)*

- On the whole, I am satisfied with myself.
- At times, I think I am no good at all.
- I feel that I have a number of good qualities.
- I am able to do things as well as most other people.
- I feel like I do not have much to be proud of.
- I certainly feel useless at times.
- I feel that I am a person of worth, at least on an equal plane with others.
- I wish I could have more respect for myself.
- All in all, I am inclined to feel that I am a failure.
- I take a positive attitude toward myself.

Appendix B Job advertisement

1. Broad information and expression of interest

   The Max-Planck-Institute in Bonn/Germany is offering a job (for a total of 4 hours) which will take place at your closest SOS program location in the next few weeks. It offers a generous hourly wage. Would you be interested in learning more?
   o I am interested in learning more.
   → *If box checked: Next page*

2. Text of the job advertisement

Please take a photo of this page or write down the e-mail address.

Job Opportunity: Assistant for the Max-Planck-Institute (4 hours)

We are looking for a reliable and organized person to help with simple office tasks at your SOS Program location. You can choose the day that works best for you! The job will comprise one or more of the following activities:

- Organizing documents and office materials
- Assisting with room setup for meetings
- Supporting various administrative tasks as needed

Details:

- Location: SOS Program location
- Date: Flexible, during May and June 2025*
- Duration: 4 hours
- Total Pay: 60 EUR**
- Travel costs covered in addition

Application: To apply, please send the following document(s) to tissen@coll.mpg.de before April 30*.

- CV (mandatory)
- Motivational letter (optional)

Your answers to the previous survey do not influence whether you get the job or not. If you're looking for a flexible, well-paid job, this is a great opportunity!
**depending on the date of the endline survey*
***in local currency*

Appendix C Additional Tables

Table 6: Number of Observations for Treatment and Control Groups by Country

Country	Control	Treatment	Intensive Treatment	N
Benin	71	51	18	140
Cape Verde	74	53	19	146
Rwanda	73	55	20	148
South Africa	82	61	21	164
Total	**300**	**220**	**78**	**598**

Notes: Number of observations per country in control, treatment and intensive treatment group in baseline. The treatment group receives 56 hours of labor market training. The intensive treatment group receives additional 8 hours of one-to-one mentoring.

Table 7: Covariates considered for randomization

Variable	**Levels**
Age	16 - 25
Male	yes/no
Is part of FS program[a]	yes/no
Married	yes/no
Has kids	yes/no
Has high school edu.[b]	yes/no
First caregiver has highschool edu.	yes/no
Ever had a job	yes/no
Looking for a job	yes/no
Monthly income[c]	in $ PPP
Part of income that is used personally[d]	1 (nothing) - 5 (everything)
Has a smartphone	yes/no
Did any labor market training before	yes/no
Network index	1 (low) - 5 (high)
Work attitude index	1 (low) - 5 (high)
Motivation index	1 (low) - 5 (high)
Skills index	1 (low) - 5 (high)
Self-esteem index	1 (low) - 5 (high)
Aggr. time pref. index[e]	0 (impatient) - 20 (patient)
Inconsistence in time pref.[f]	yes/no

Notes: Variables in this table were used for the minMSE algorithm (Schneider and Schlather, 2017) for randomization. Index variables are weighted means of different survey questions. a) Family strengthening. b) 25 youth (4% of the whole sample) have no highschool-degree, are below 18 and are still in school. c) Most participants (69%) have no income. d) For most participants, this is hypothetically. e) Combines answers of trade-off between "now" and "in one week" and between "in one week" and "in two weeks". f) Dummy, whether the number of switches between receiving amount earlier and later in one decision table is above median or not.

Table 8: Economic Indicators per country

	Youth unemployment (15-24, in %)	Informality (in %)	GDP Growth 2025 (in %)	Population with Highschool Degree (in %)
Benin	3.2	99.8	6.5	12.52
Cape Verde	61.6	83.8	4.7	55.65
Rwanda	18.3	95	6.5	20.7
South Africa	60.1	45.3	1.5	61.46

Notes: Descriptive statistics for the four countries, Informality according to the ILO definition. Expect higher (lower) employment at endline for green (red) numbers. *Source: ILOSTAT* for unemployment and informality, *IMF* for projected GDP growth and *Unesco Institute for Statistics* for education.

Table 9: Labor market potential by location

	Population Size	Dist. large city	Restaurants	Market size	Baseline employ.
Benin					
Abomey-Calavi	500K+	0 - 50 Km	10 - 20	High	0.25
Dassa-Zoume	0 - 50K	50 - 200 Km	20 - 100	Medium	0.07
Natitingou	50K - 100K	200+ Km	20 - 100	Medium	0.30
Cape Verde					
Praia	100K - 500K	200+ Km	20 - 100	Medium	0.20
Sao Felipe	0 - 50K	200+ Km	10 - 20	Low	0.18
Rwanda					
Byumba	50K - 100K	0 - 50 Km	0 - 10	Medium	0.30
Kayonza	0 - 50K	50 - 200 Km	0 - 10	Low	0.03
Kigali	500K+	0 - 50 Km	100+	Very high	0.33
South Africa					
Cape Town	500K+	0 - 50 Km	100+	Very high	0
Ennerdale/ Mamelodi	100K - 500K	0 - 50 Km	20 - 100	High	0
Mathanjana	0 - 50K	50 - 200 Km	0 - 10	Low	0.04
Nelspruit	100K - 500K	200+ Km	100+	High	0.04
Qwa Qwa (Makeneng)	50K - 100K	200+ Km	0 - 10	Low	0
Rustenburg	500K+	0 - 50 Km	100+	Very high	0
Sekhukhune	0 - 50K	50 - 200 Km	10 - 20	Low	0

Notes: Statistics on the program locations corresponding to the 15 randomization strata. Population sizes are projections for 2025; the last census data was used if the projection was not available. Large city if population size is at least 500 thousand. The potential labor market size is higher if the population in the city/town of the location is larger, if a city of at least 500 thousand inhabitants is closer, and if there are more restaurants. *Source:* Census data and *World Population Review* for population, *Google Maps* for distance calculations, *Google Maps* and *Tripadvisor* for number of restaurants, and baseline survey for youth employment.

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